

Banking Risk & Loan Performance Analytics

Interview Notes — Concept Notes, Project Story, Business Logic, KPIs, DAX, SQL Thinking and Common Mistakes

Use this as revision notes. It is intentionally written without a question-answer format. Read the notes until you can explain the project naturally.

1. Project at a Glance

This project is an end-to-end **descriptive banking analytics** project. The workflow is MySQL for structured storage, Python/Pandas for exploratory analysis, and Power BI/DAX for interactive reporting.

Business focus

- Understand the size and composition of the lending portfolio.
- Understand the size and composition of the deposit portfolio.
- Compare lending and deposits across customer segments.
- Understand banking relationships, customer engagement, income bands, nationality, loyalty classification, and advisor relationships.
- Identify patterns that may deserve deeper risk investigation.

Project scope

The current implementation is **descriptive and exploratory**. It is not a machine-learning credit-risk model. It does not predict repayment and does not automatically approve or reject loans.

Natural project story

Raw banking data → MySQL → Python EDA → Power BI model → DAX → 5-page dashboard → portfolio insights and areas for further investigation.

2. Banking Business Basics

A bank receives customer money through deposits and provides money through loans and credit products. The bank earns income through interest and fees while managing the possibility of loss when borrowers do not repay.

Simple banking picture

Customers → Deposits / Accounts → Bank → Loans / Credit → Repayment + Interest + Fees

The two sides in this project

Side	Meaning	Your measures
Lending	Money outstanding through credit products.	Bank Loan, Business Lending, Credit Card Balance, Total Loan
Deposits	Customer funds held in bank products.	Bank Deposit, Savings, Checking, Foreign Currency, Total Deposit
Client relationship	Customer size and relationship context.	Total Clients, Engagement Length
Fees	Estimated fee value based on the project rule.	Total Fees, Processing Fees

3. Banking Vocabulary

Deposit — Money a customer keeps with the bank.

Loan — Money lent by the bank that the borrower is expected to repay.

Business Lending — Credit provided for business use.

Credit Card Balance — Outstanding borrowing associated with a credit card.

Interest — A charge paid by a borrower or a return paid to a depositor, depending on context.

Fee — A charge collected for a product or service.

Processing Fee — A charge for processing a product/loan; your project maps the fee structure to a percentage.

Portfolio — The complete group of loans, deposits, or products being analyzed.

Exposure — Amount outstanding or at risk under a defined business rule.

Credit Risk — Possibility that a borrower fails to meet repayment obligations.

Delinquency — A payment is past due under a defined rule.

Default — Failure to meet a contractual repayment obligation.

Risk Segmentation — Grouping customers/loans using relevant characteristics for monitoring.

Concentration — A large share of value depends on one customer group, product, relationship, or segment.

Reconciliation — Checking that related totals or records agree.

Data Lineage — Tracing a reported value back to its source and transformations.

Access Control — Rules defining who can access or modify data.

4. Data and MySQL Layer

Database: **banking_risk**

Tables used:

- customer_table
- banking-relationships
- gender
- investment-advisors

Why use MySQL?

MySQL provides a structured database layer where related banking tables can be stored and queried. It separates the data-storage/SQL layer from the Python analysis and Power BI reporting layer.

Data-model thinking

The customer table is the main customer-level table containing lending, deposits, income, fee structure, loyalty, engagement and account information. The related tables provide additional attributes such as banking relationship, gender and investment advisor.

5. Python / Pandas EDA

The EDA stage is used to understand the dataset before building the final dashboard. It looks at numerical variables, categorical variables, distributions, segment patterns and relationships between banking measures.

Why EDA matters

- It helps identify unusual values and distribution shapes.
- It helps understand which variables are useful for business segmentation.
- It helps determine which patterns are worth carrying into the dashboard.
- It provides an independent check before visualization.

Descriptive analysis principle

Because this project is not predictive modeling, there is no requirement to transform every skewed numerical variable. A skewed distribution is an EDA finding. Transformation should only be introduced when there is a specific analytical reason.

6. Power BI Calculated Columns

Engagement Days

Measures the number of days from Joined Bank to TODAY(). It gives a simple view of how long the customer has been associated with the bank.

Income Band

Groups estimated income into Low, Mid and High based on the thresholds implemented in the project.

Processing Fees

Maps the Fee Structure to rates: High = 5%, Mid = 3%, Low = 1%.

Engagement Timeframe

Converts engagement days into readable groups: <1 year, <5 years, <10 years, <20 years, and >20 years.

7. DAX Measure Logic

Measure	Meaning
Total Clients	Distinct count of Client ID.
Bank Deposit	Sum of Bank Deposits.
Bank Loan	Sum of Bank Loans.
Business Lending	Sum of Business Lending.
Checking Accounts	Sum of Checking Accounts.
Credit Cards Balance	Sum of Credit Card Balance.
Savings Account	Sum of Saving Accounts.
Foreign Currency Account	Sum of Foreign Currency Account.
Total CC Amount	Sum of Amount of Credit Cards.
Total Deposit	Bank Deposit + Savings + Foreign Currency + Checking.
Total Loan	Bank Loan + Business Lending + Credit Card Balance.
Total Fees	SUMX of Total Loan $\times$ Processing Fees for each customer row.
Engagement Length	Sum of Engagement Days.

Why SUMX is important

Total Fees is a row-level calculation because the processing rate can differ by customer. SUMX evaluates the expression customer by customer and then adds the results.

8. Dashboard Story

Page 1 — Executive Overview

Purpose: quickly understand client volume and overall lending, deposits, fees, credit-card amount and savings.

Page 2 — Lending & Loan Analysis

Purpose: understand the lending portfolio across banking relationship, gender, investment advisor, engagement timeframe, income band and nationality.

Main visuals: Bank Loan by Banking Relationship; Lending Portfolio by Engagement; Bank Loan by Income Band; Bank Loan by Nationality.

Page 3 — Deposits & Funding Analysis

Purpose: understand deposits and deposit composition across engagement timeframe, nationality and income band.

Main visuals: Bank Deposit by Engagement Timeframe; Deposit Portfolio by Nationality; Bank Deposits by Income Band.

Page 4 — Banking Portfolio Summary

Purpose: provide a consolidated view of the main portfolio KPIs.

Page 5 — Client Details

Purpose: move from portfolio reporting to client-level information, fee analysis by loyalty classification, and fee/engagement comparison by nationality.

9. Key Project Numbers

Metric	Project value
Total Clients	2,940
Total Loan	\$4.38bn
Total Deposit	\$3.77bn
Total Fees	\$158.19M
Bank Loan	\$1.77bn
Business Lending	\$2.60bn
Savings Accounts	\$698.73M
Checking Accounts	\$963.28M
Foreign Currency	\$89.65M

These are project-dataset outputs. Do not present them as real-world banking benchmarks.

10. Business Problem → Analysis → Solution

Problem: Lending portfolio is large.

Analysis: split Total Loan into Bank Loan, Business Lending and Credit Card Balance, then segment by relationship, income band, nationality and engagement timeframe.

Solution: identify where the largest lending value is concentrated and flag segments for deeper risk investigation.

Problem: Management needs to understand deposits.

Analysis: split Total Deposit into Bank Deposit, Savings, Checking and Foreign Currency, then compare by nationality, income band and engagement timeframe.

Solution: make the deposit mix and major contributing segments visible.

Problem: Customer fee value varies.

Analysis: calculate processing fees at customer level and summarize them by loyalty classification and other segments.

Solution: identify where fee value is concentrated and which customer groups may be commercially important.

11. Credit Risk Thinking

Credit risk means the possibility that a borrower does not meet repayment obligations and the bank suffers a loss.

A portfolio question is different from a credit-risk prediction question.

Portfolio question: **Where is lending concentrated?**

Risk question: **How likely is that exposure to create a loss, and how large could that loss be?**

The current project supports the first question much better than the second. A true credit-risk analysis would normally need repayment history, days past due, default status, risk grade/score, outstanding principal, loan terms, and other validated risk information.

Safe interpretation of current dashboard

- Use customer attributes for segmentation.
- Use portfolio concentration as a signal for further investigation.
- Do not call a segment “high risk” without appropriate risk data.
- Do not claim nationality, gender, income, or engagement directly determines creditworthiness.

12. Business Questions to Practice Mentally

Use these as thinking prompts, not memorized answers. The key is to know the analysis path.

- Where is the largest lending exposure?
- Which banking relationship contributes the most Bank Loan?
- Which income band contributes the most lending?
- How is lending distributed by nationality?
- How does lending differ across engagement timeframes?
- Which deposit type contributes the most to Total Deposit?
- Which income band contributes the most deposits?
- How do deposits differ across nationalities?
- Which loyalty class contributes more fee value?
- Which advisor is associated with higher portfolio values?
- Which segments have high lending but would need more risk information before a decision?
- How would you investigate an unusual concentration?

General solution pattern

Define → measure → validate → segment → compare → prioritize → investigate → recommend → monitor.

13. SQL Thinking for the Project

Typical business-to-SQL translations:

Business requirement	SQL approach
Count clients	COUNT(DISTINCT Client_ID)
Total bank loan	SUM(Bank_Loans)
Total business lending	SUM(Business_Lending)
Loan by relationship	GROUP BY Banking_Relationship
Loan by income band	CASE + GROUP BY
Deposit by nationality	GROUP BY Nationality
Top advisors by loan	GROUP BY Advisor + ORDER BY Loan DESC
Fees by loyalty class	GROUP BY Loyalty_Classification
Average loan per customer	Total loan / distinct client count

The important skill is not memorizing one query. It is converting a business question into the correct aggregation, grouping, filter, and comparison.

14. Data Quality and Governance Thinking

Before using a banking report for decisions, an analyst should think about:

- Missing or invalid customer identifiers.
- Incorrect or impossible numeric values.
- Duplicate records.
- Incorrect relationship keys.
- Missing or inconsistent dates.
- Reconciliation with the authoritative source.
- Clear KPI definitions.
- Data lineage and traceability.
- Access controls for customer-level information.
- Privacy and appropriate use of demographic attributes.

A reliable banking number must be explainable: where it came from, how it was transformed, what definition was used, and what limitations remain.

15. Common Mistakes to Avoid

- Calling the project a predictive loan-repayment model when no predictive model was built.
- Calling a customer or demographic group high-risk based only on descriptive attributes.
- Presenting portfolio concentration as proof of risk.
- Using portfolio totals without considering customer counts.
- Ignoring the metric definition and denominator.
- Skipping data validation and reconciliation.
- Making recommendations that are too broad, such as “reduce risk” without a specific next step.
- Claiming Workday, Qlik, production risk modeling, or regulatory expertise that was not actually used.
- Adding advanced banking metrics to the resume that were not calculated in the project.

16. Final Interview Story

“This project focuses on banking portfolio analysis. I used MySQL to store the banking tables, Python and Pandas for exploratory analysis, and Power BI with DAX for the final reporting layer. I created customer segments such as income band and engagement timeframe and measures for total clients, loans, deposits, and fees. The dashboard is organized into executive overview, lending, deposits, portfolio summary, and client details. The analysis helps identify how lending and deposits are distributed across customer segments and relationships. It is a descriptive project, so I use the results to identify patterns and areas for further risk investigation rather than claiming that the dashboard predicts repayment or determines credit approval.”

Final mental model

Banking business → **customer/portfolio data** → **KPI definition** → **segmentation** → **pattern** → **investigation** → **business action**.

The goal for the interview is to sound like someone who understands both the **business problem** and the **technical work**, not someone who only remembers the dashboard.